

2.3 One Quantitative Variable: Using Percentiles

- Goal: We need to turn “raw” data into information.
- Data are summarized into
 - 1 tables & graphs
 - 2 numerical indices
- One quantitative variable
 - 1 Five-number summary
 - 2 Percentiles

Three Features of Data

■ What look like the data?

- 1 The **shape** of the distribution of the data.
- 2 While there are many different variety of shapes, certain shapes are common enough to have been named.

■ Where is the center of the data?

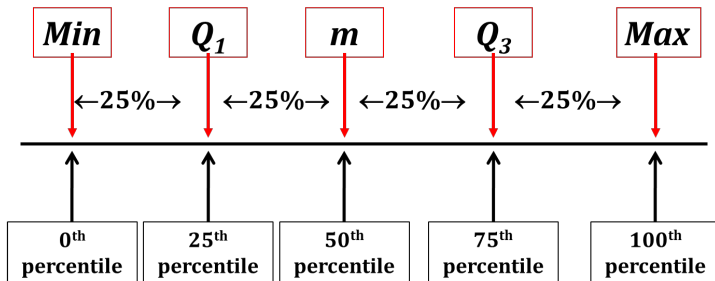
- 1 The **central location** of the distribution of the data is the most important aspect to notice and describe.
- 2 Measures of central location: **Mean**, **Median**, **Mode**.

■ How spread out the data?

- 1 The **variability** of the distribution of the data is the second most important feature.
- 2 Measures of spread: **Standard deviation**, **IQR**, **Range**.

Percentiles and Quartiles

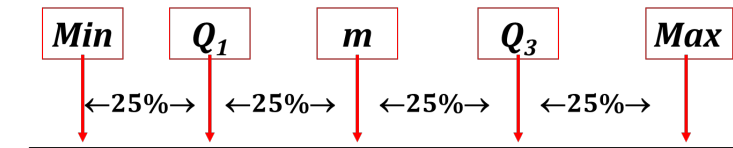
- The **P th percentile** is the value that, after the data are arranged in the **ascending** order, has **$P\%$ of the data below** it and **$(100 - P)\%$ above** it.
 - 1 **25th percentile** is also called the **first quartile** (or **lower quartile**) and denoted by Q_1 .
 - 2 **50th percentile** is also called the **second quartile** (or **middle quartile**) and denoted by Q_2 , which is also the **median m** .
 - 3 **75th percentile** is also called the **third quartile** (or **upper quartile**) and denoted by Q_3 .



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Finding Quartiles

- A “shortcut” method for finding lower and upper quartiles:
 - 1 Rearrange the data from the smallest to the largest.
 - 2 Divide the rearranged data into two subsets using the median m as the dividing value.
 - 3 Q_1 = the median of the values below the median m .
 - 4 Q_3 = the median of the values above the median m .



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

- There are several different ways of computing percentiles and quartiles.

IQR and Range | Measure of Spread

- The **interquartile range (IQR)** is the distance between Q_1 and Q_3 :

$$\text{IQR} = Q_3 - Q_1$$

- The **range** is the largest value minus the smallest value:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

- Notes

- 1 IQR is the **range** of the **middle 50%** data values.
- 2 IQR is **less sensitive to outliers** than any other measures of variability because it completely ignores the extremes in the data.
- 3 IQR can be very misleading when the data set is highly concentrated about the median. For instance, the data sets A and E.

Example 2.12 | IQR

- Compute three quartiles and IQR for the following data set.

2	3	2	6	4	1	5	6	1
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- ① To find three quartiles, we first arrange the data values in ascending order as follows:

1	1	2	2	3	4	5	6	6
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- ② The median is $m = 3$, and this median divides the data into the smaller half (1, 1, 2, 2) and the larger half (4, 5, 6, 6).
- ③ We find $Q_1 = 1.5$, which is the median of the smaller subset (1, 1, 2, 2).
- ④ We obtain $Q_3 = 5.5$, the median of the larger subset (4, 5, 6, 6).
- ⑤ Therefore, we compute

$$\text{IQR} = Q_3 - Q_1 = 5.5 - 1.5 = 4.0$$

Exercise 2.6 | IQR

- Consider the following data sets:

Data set A	10	10	10	10	10	10	$\bar{x} = 10, m = 10$
Data set B	5	5	5	15	15	15	$\bar{x} = 10, m = 10$
Data set C	0	0	0	20	20	20	$\bar{x} = 10, m = 10$
Data set D	7	8	9	11	12	13	$\bar{x} = 10, m = 10$
Data set E	0	10	10	10	10	20	$\bar{x} = 10, m = 10$

Compute the IQRs for the five data sets.

Data set	Q_1	Q_3	IQR	Rank
A				
B				
C				
D				
E				

Five-number Summary

- The **five-number summary** is composed of
 - 1 Min (the smallest value)
 - 2 Q_1 (the first quartile)
 - 3 m (the median)
 - 4 Q_3 (the third quartile)
 - 5 Max (the largest value)

Example 2.12 (revisited) | Five-number summary

- Find the five-number summary for the following data set.

2	3	2	6	4	1	5	6	1
---	---	---	---	---	---	---	---	---

- To find three quartiles, we first arrange the data values in ascending order as follows:

1	1	2	2	3	4	5	6	6
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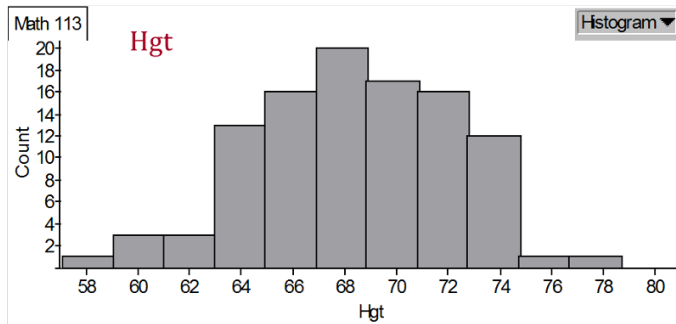
- The median is $m = 3$, and this median divides the data into the smaller half (1, 1, 2, 2) and the larger half (4, 5, 6, 6).
- We find $Q_1 = 1.5$, which is the median of the smaller subset (1, 1, 2, 2).
- We obtain $Q_3 = 5.5$, the median of the larger subset (4, 5, 6, 6).
- The five-number summary is

Min	Q_1	$\tilde{x}$	Q_3	Max
1	1.5	3	5.5	6

Exercise 2.7 | Five-number summary

■ Based on this histogram of heights, estimate the following:

- 1 The mean
- 2 The standard deviation
- 3 The five-number summary
- 4 The 10th percentile



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Detection of Outliers



Summary | One Quantitative Variable

- Visualizing one quantitative variable:
 - 1 Dotplot
 - 2 Frequency (or relative frequency) table
 - 3 Histogram
- Shape: symmetric, bell-shaped, skewed
- Summary Statistics
 - 1 Measures of center: mean (not resistant to outliers), median (resistant to outliers)
 - 2 Measures of spread: standard deviation (not resistant to outliers), IQR (resistant to outliers), range (not resistant to outliers)
 - 3 Percentiles
 - 4 Five-number summary: min, Q_1 , median, Q_3 , max
- Other concepts
 - 1 Outliers, resistance
 - 2 z-scores